

The Effects of Weather Conditions On Energy Generation and Consumption

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Context, Motivation & Study Overview

- Renewable energy reliability is challenged by strong weather dependency
- Solar and wind generation are inherently weather-dependent
- Weather-driven variability complicates supply–demand balance and forecasting
- Spain’s hourly energy generation, demand, price, and weather data (2015–2018)
- Statistical hypothesis testing and machine learning methods are applied
- Temperature and weather conditions strongly influence demand and solar output

RESEARCH QUESTIONS

1. *What are the hourly, daily, and seasonal patterns of energy consumption and generation?*
2. *How does wind speed affect wind-power production?*
3. *Do different weather conditions (e.g., clouds, rain, clearsky) significantly influence the amount of solar energy generated?*
4. *Does energy demand fluctuate significantly during temperature extremes (high vs. low) compared to moderate conditions?*
5. *Does temperature variability lead to increased volatility (standard deviation) in energy demand?*
6. *Does energy generation exhibit statistically significant fluctuations across the four seasons?*
7. *To what extent can meteorological variables (humidity, wind speed, temperature) linearly predict total energy load?*
8. *Can weather data be used to classify energy demand levels (High vs. Low) with accuracy?*
9. *Can advanced machine learning models (Gradient Boosting) outperform traditional statistical baselines (SARIMAX) in forecasting grid metrics?*
10. *Is it possible to accurately forecast electricity prices using only available data without future weather inputs?*

DATASET & KEY VARIABLES

- Hourly energy generation, demand, and electricity prices (Spain)
- Hourly meteorological observations
- Study period: 2015–2018
- Energy and weather data merged by timestamp

Key Energy Variables

- Total load (demand)
- Solar, wind, fossil, and nuclear generation
- Electricity price

Key Weather Variables

- Temperature
- Wind speed
- Humidity
- Cloud cover
- Weather description

DESCRIPTIVE ANALYSIS

Electricity Price Distribution

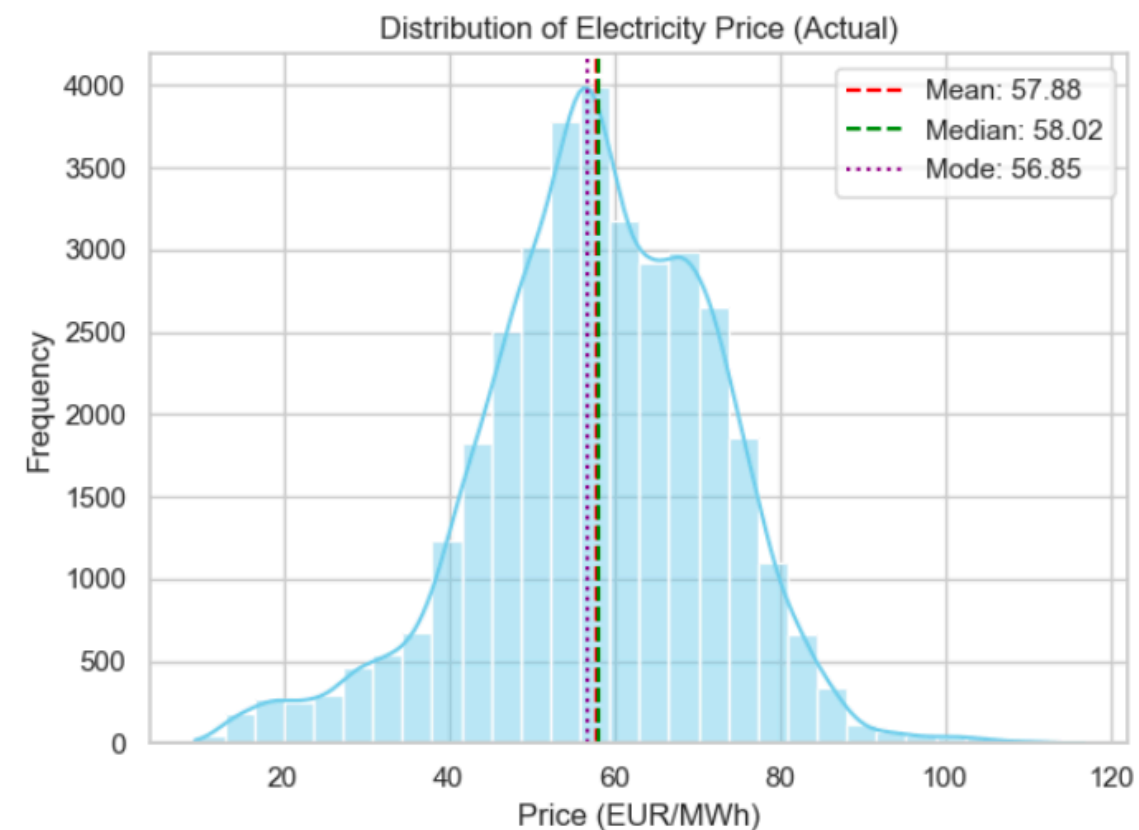


Fig. 1. Distribution of Actual Electricity Prices (EUR/MWh)

- Prices centered around ~58 EUR/MWh
- Approximately normal distribution
- Occasional spikes suggest stress periods

Electricity Prices Over Time

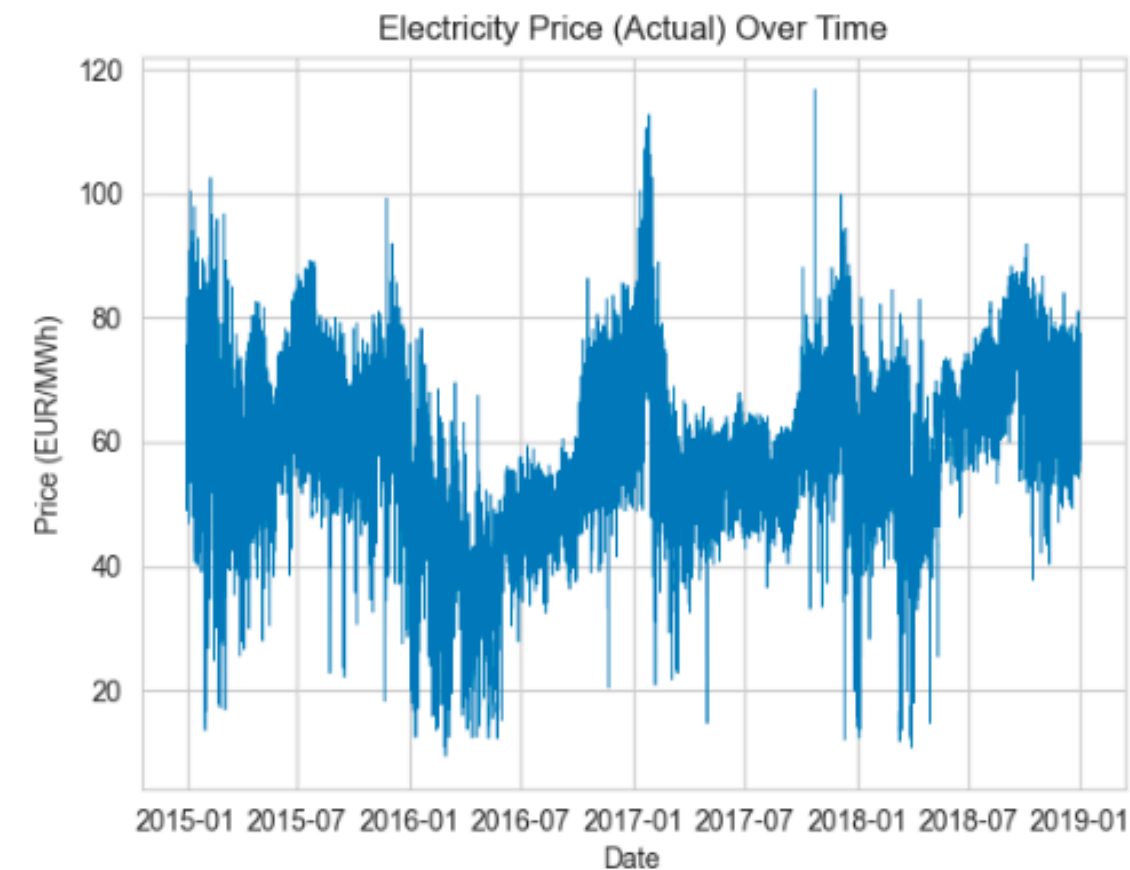
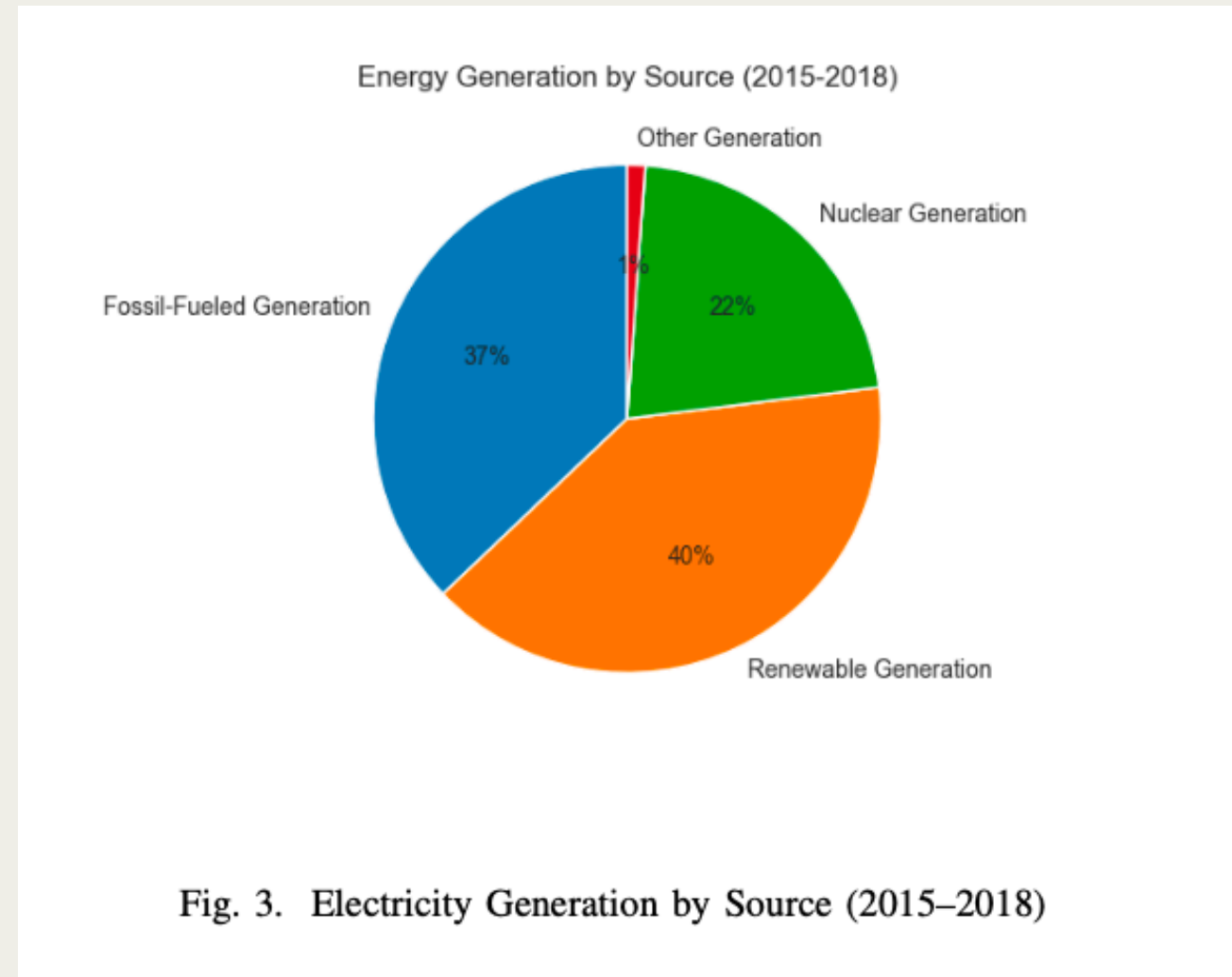


Fig. 2. Actual Electricity Price Over Time (2015–2018)

- Seasonal price increases in summer and winter
- Higher prices align with peak demand periods
- Short-term spikes indicate external stress factors

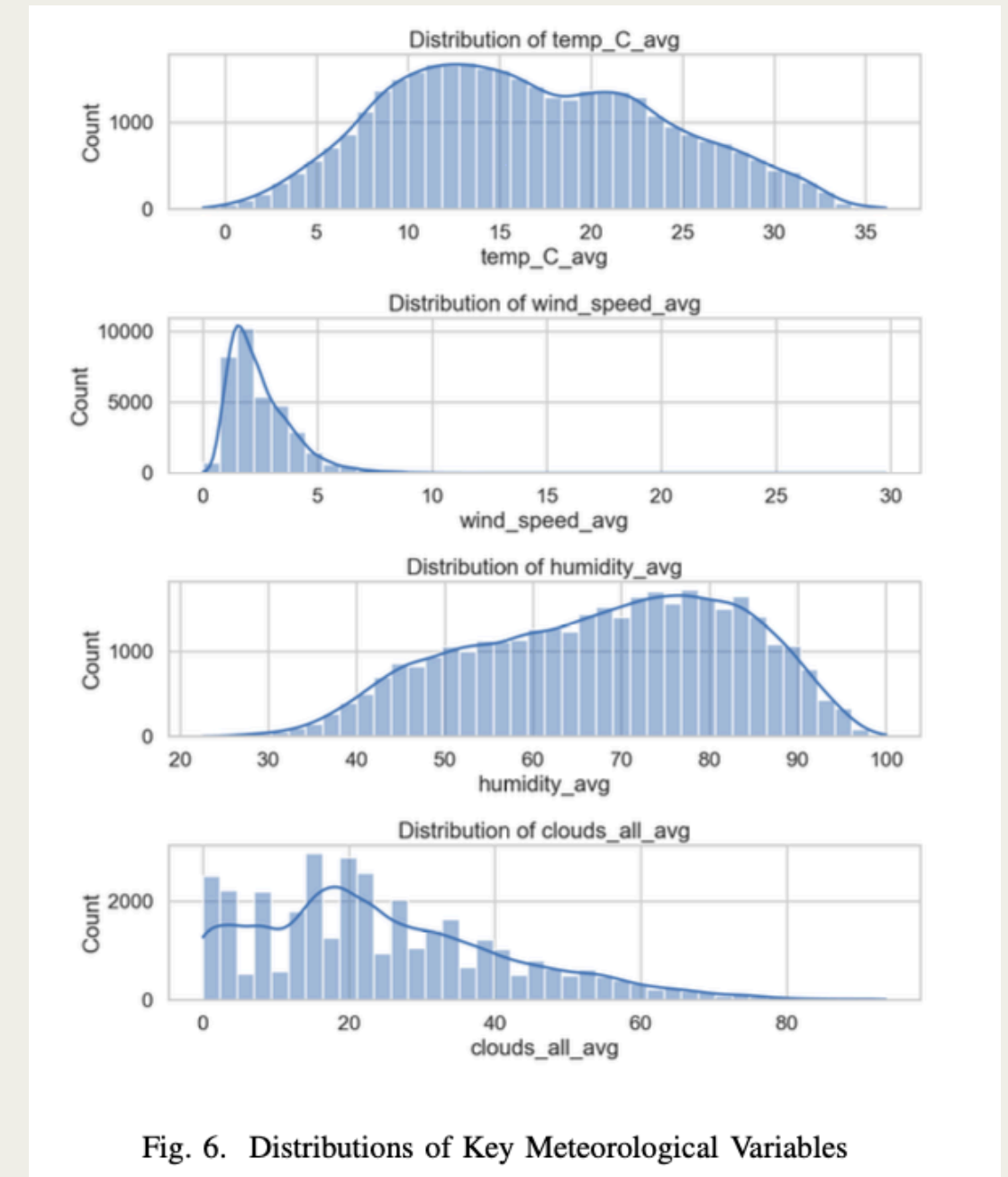
DESCRIPTIVE ANALYSIS

Energy Generation Mix



- Renewable energy accounts for ~40% of generation
- Fossil fuels ~37%, nuclear ~22%
- Diversified generation mix stabilizes the system

Distribution of Weather Variables



- Temperature shows clear seasonal structure
- Wind speed is right-skewed, mostly moderate
- High humidity and variable cloud cover affect solar output

DESCRIPTIVE ANALYSIS

Weather Conditions Overview

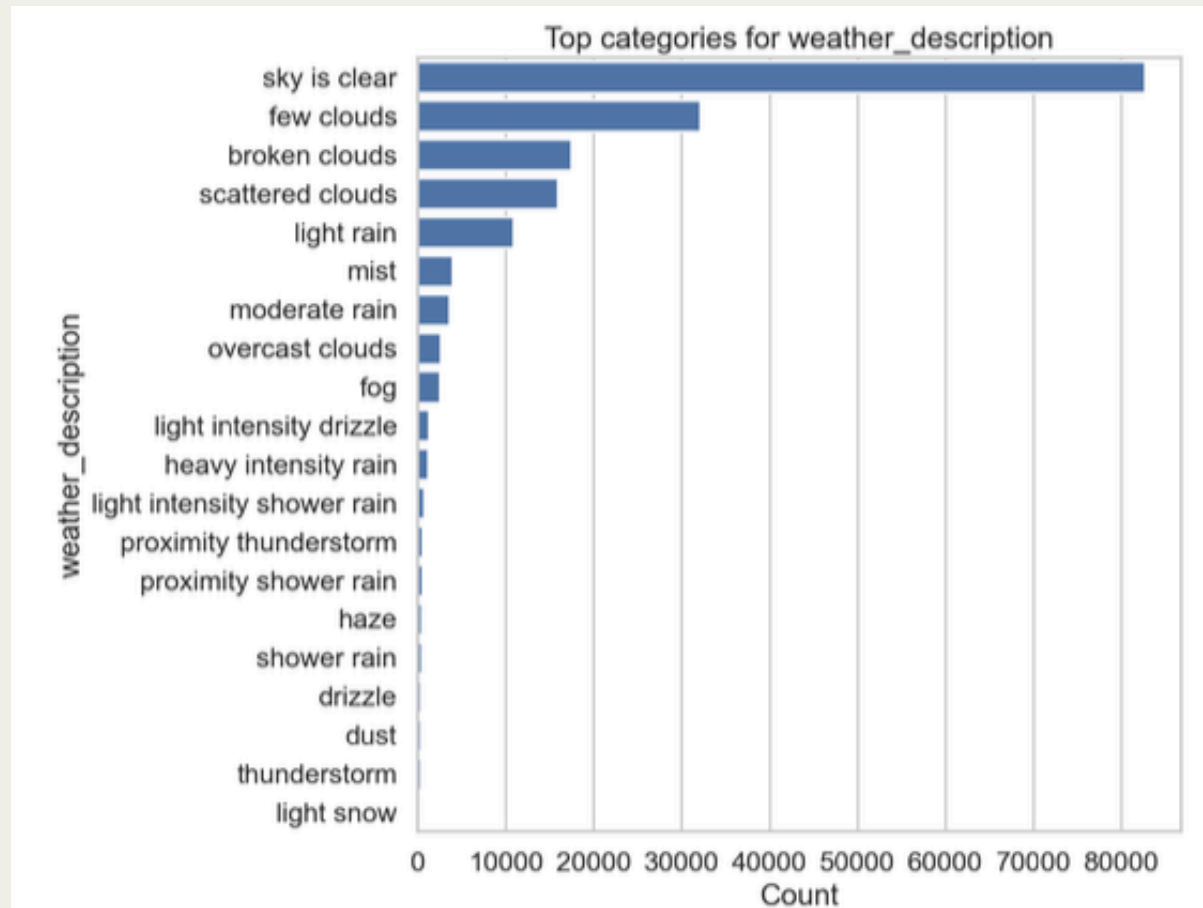


Fig. 4. Frequencies of Weather Descriptions

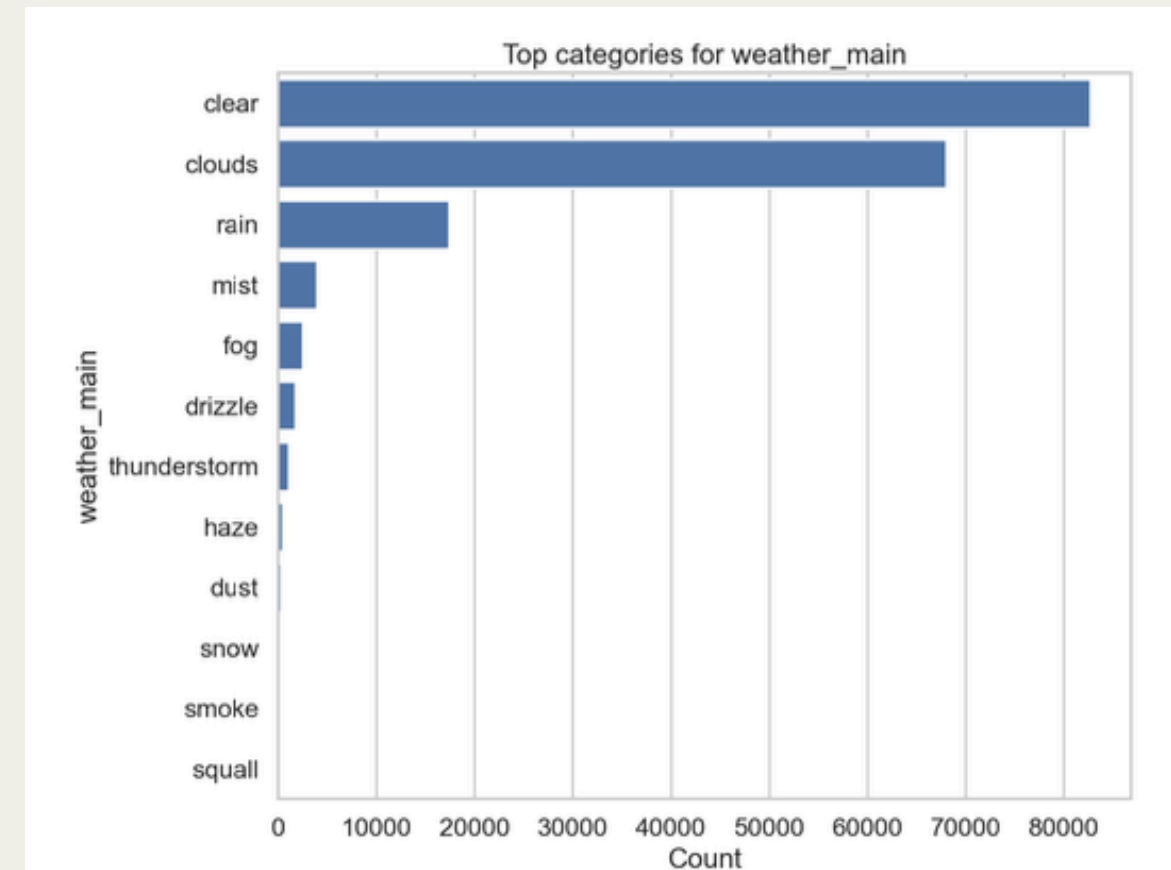


Fig. 5. Frequencies of Weather Main Categories

- Clear sky is the most frequent condition
- Cloudy conditions dominate remaining observations
- Relevant for evaluating solar generation performance

DESCRIPTIVE ANALYSIS

Energy Demand Patterns

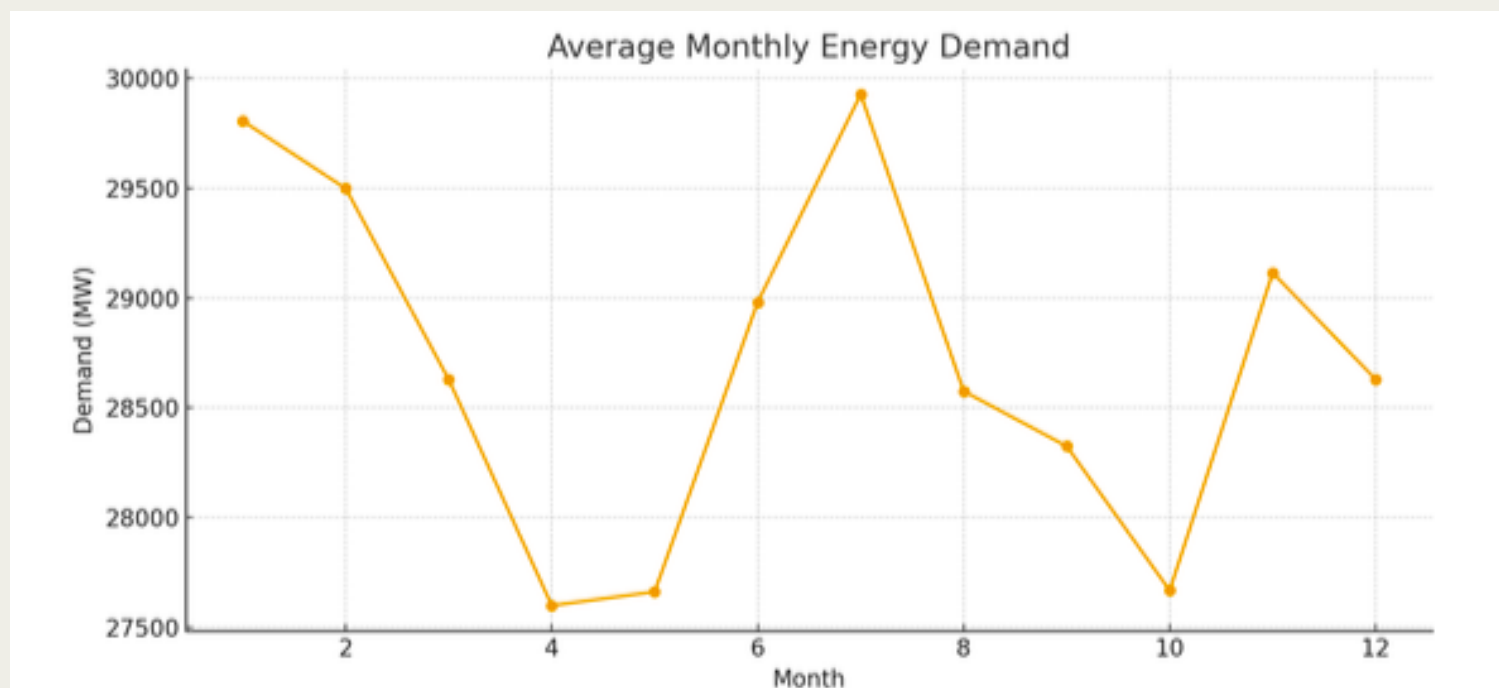


Fig. 7. Monthly Average Energy Demand (2015–2018)

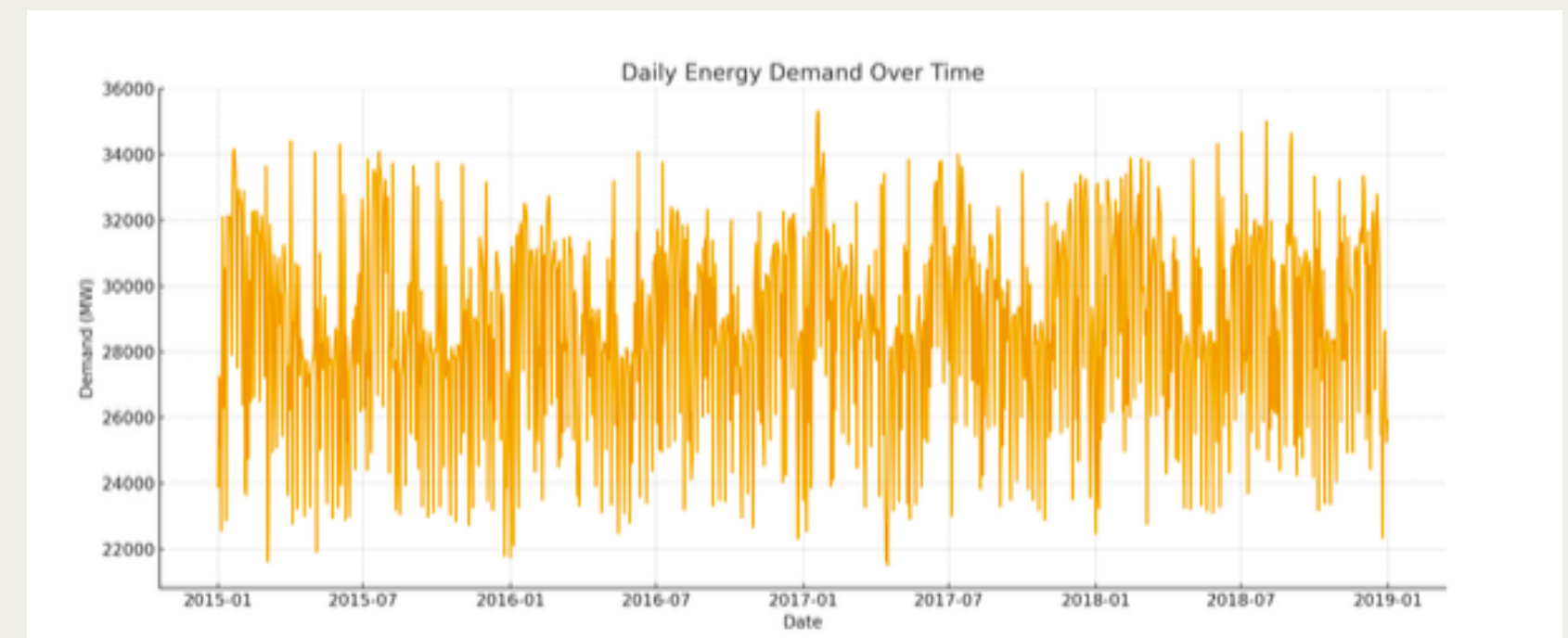


Fig. 8. Hourly Average Energy Demand (2015–2018)

- Demand peaks in winter and summer
- Extreme temperatures increase electricity consumption
- Clear link between demand and price dynamics

DESCRIPTIVE ANALYSIS

Solar Generation Under Different Weather Conditions

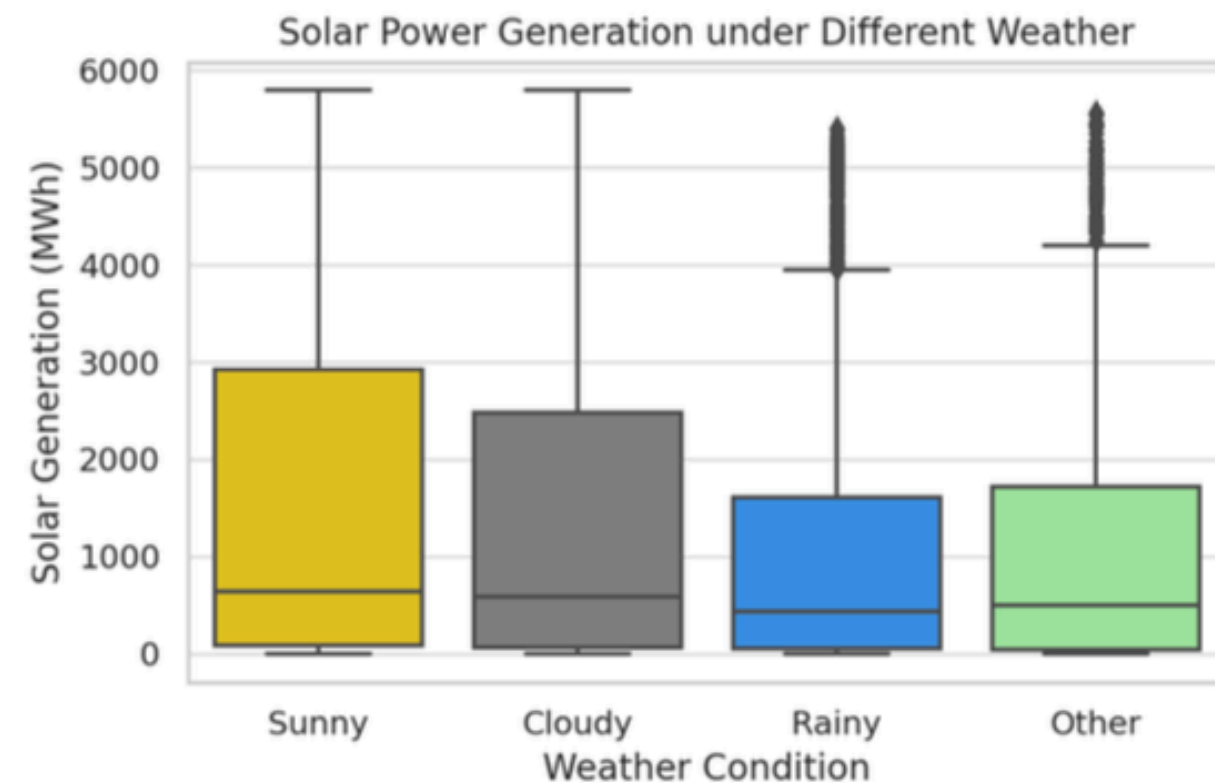


Fig. 9. Solar Power Generation Across Weather Conditions

- Solar output is highest during clear weather
- Cloudy and rainy conditions significantly reduce production
- Solar generation remains non-zero even under adverse conditions

DESCRIPTIVE ANALYSIS

Energy Demand Patterns

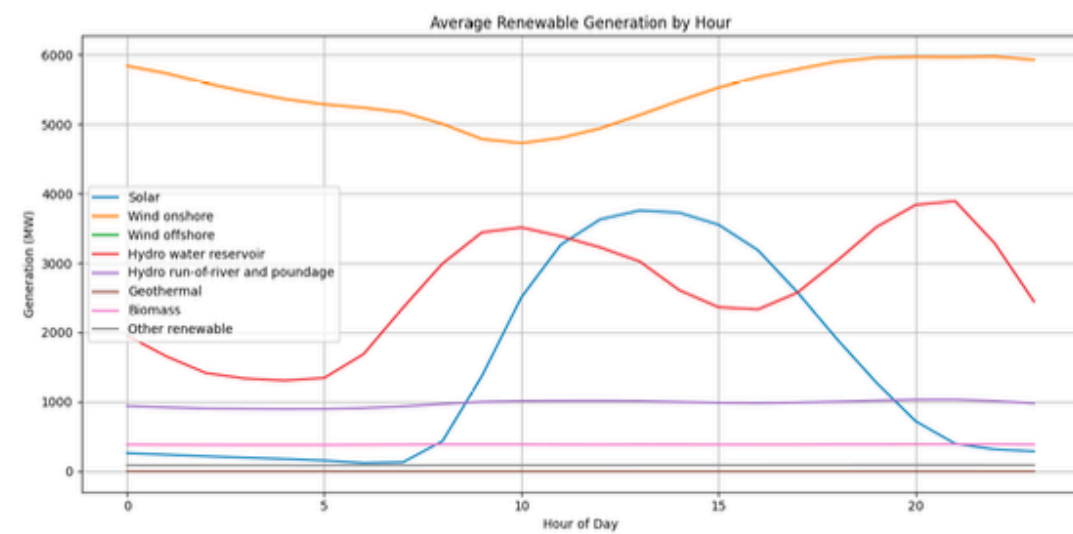
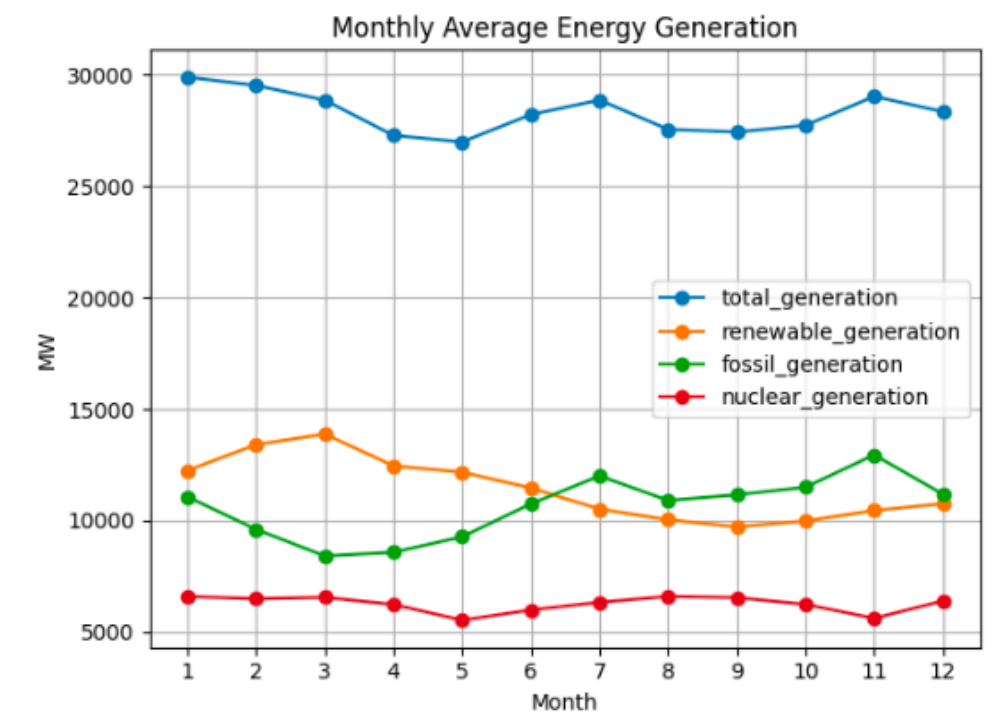
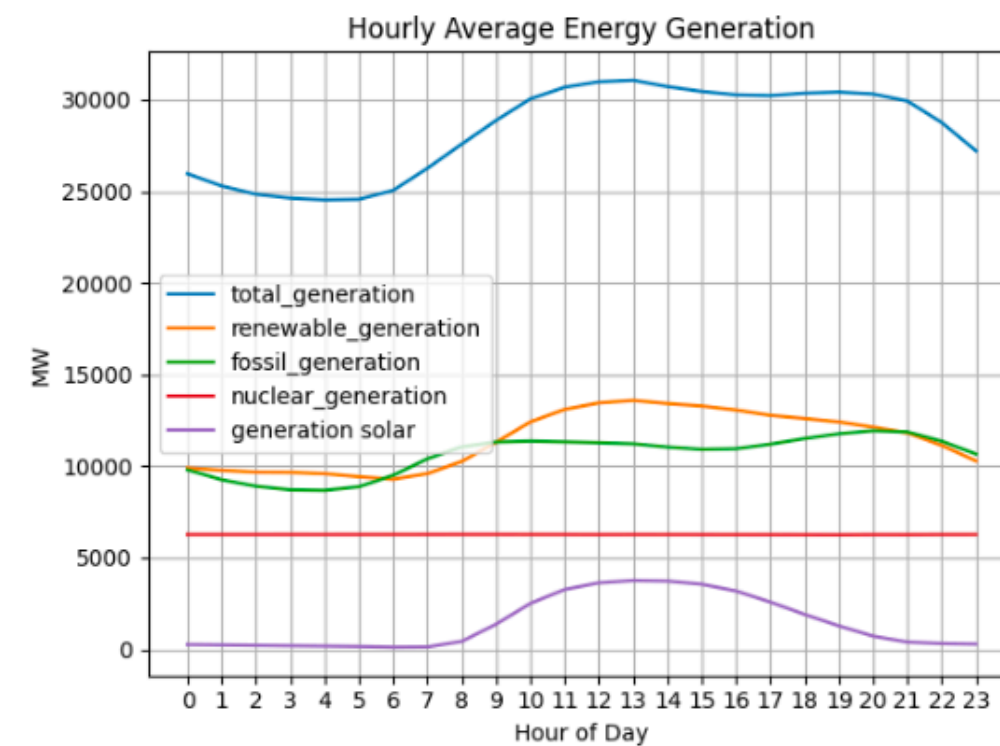
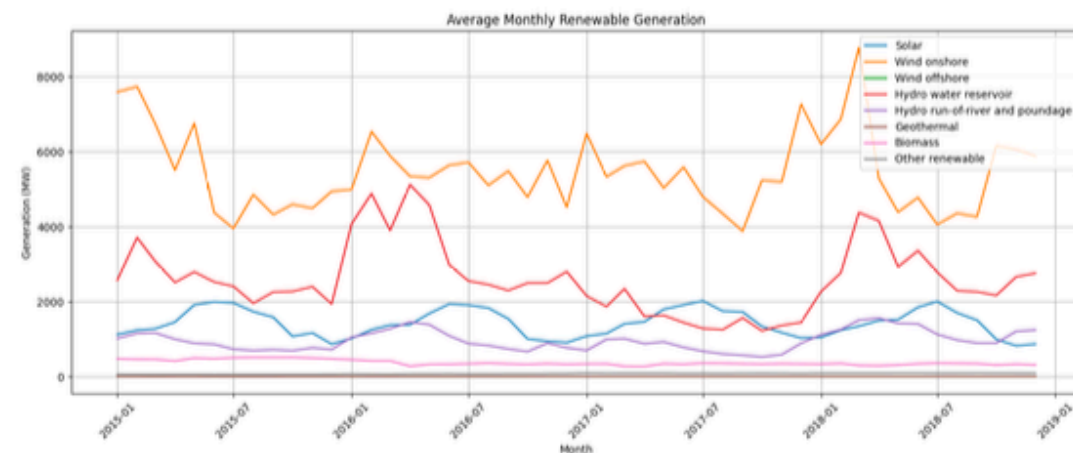


Fig. 12. Hourly Average Renewable Generation by Type (2015–2018)



- Nuclear generation remains stable
- Renewable generation peaks midday
- Fossil generation compensates during low renewable output

STATISTICAL ANALYSIS

Weather Impact on Solar Power

Do different weather conditions significantly influence the amount of solar energy generated?

- H0: Weather has no effect on solar generation.
- H1: At least one type of weather condition affects solar generation in a statistically significant way
- Solar generation varies substantially across weather categories
- Highest mean output: Snow (2751 MW) and Fog (1787 MW)
- Lowest mean output: Drizzle (604 MW) and Clouds (765 MW)
- High variability observed across all conditions (Std. Dev. $\approx$ 800–1700 MW)
- Shapiro–Wilk test indicates non-normal distributions in most groups
 - $p < 0.05$ for all weather types except Snow ($p = 0.0897$)
- One-Way ANOVA: $F = 7.30$, $p < 0.0001$
- Kruskal–Wallis: $H = 46.83$, $p < 0.0001$
- H0 rejected: Weather conditions have a statistically significant effect on solar generation
- Clouds and drizzle cause the strongest reduction in solar output

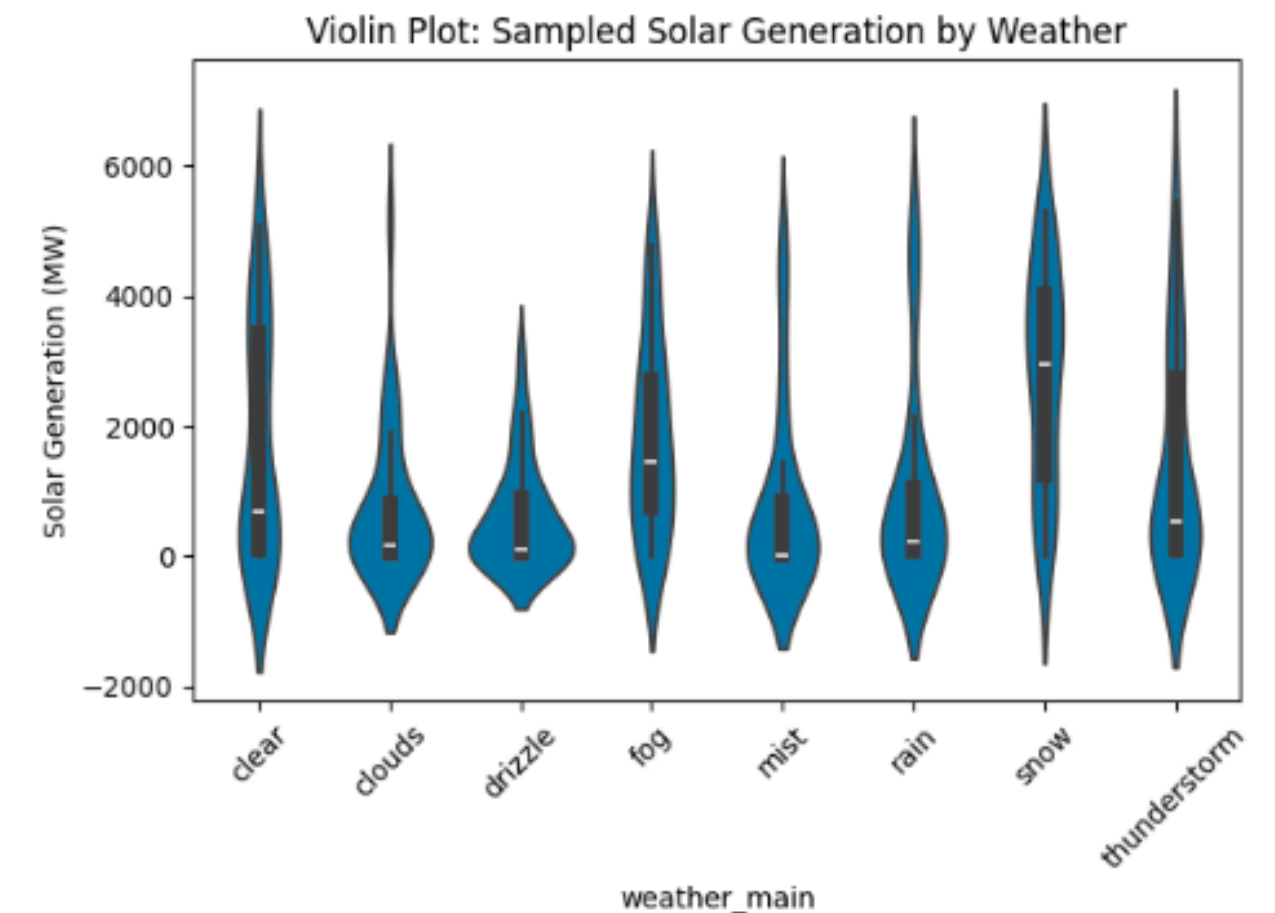
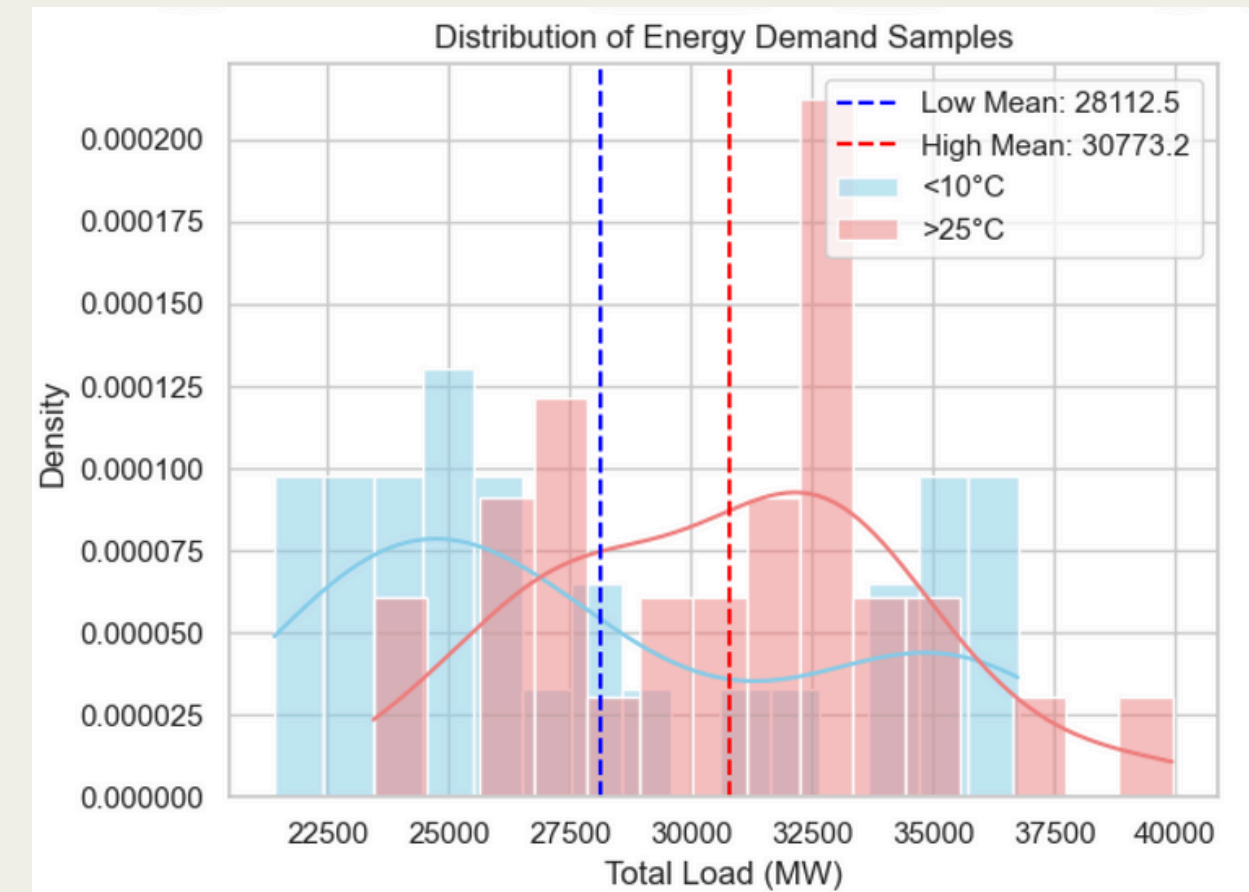


Fig. 16. Solar Power Generation Across Weather Conditions

STATISTICAL ANALYSIS

Relationship Between Temperature Extremes and Energy Demand

- Does energy demand fluctuate significantly during temperature extremes (high vs. low temperatures)?
- H0: Energy demand remains constant regardless of temperature extremes.
- H1: There is a significant difference in energy demand between low-temperature and high-temperature periods.
- Energy demand analyzed under two extreme temperature bands
- Low Temp: $< 10^{\circ}\text{C}$ ($n = 7205$)
- High Temp: $> 25^{\circ}\text{C}$ ($n = 5078$)
- Mean demand increases at high temperatures
- Low Temp: 28,113 MW
- High Temp: 30,773 MW
- Demand variability differs across temperature extremes
- Low Temp Std. Dev.: 5,112 MW
- High Temp Std. Dev.: 3,832 MW
- Statistically significant difference in mean demand
- H0 rejected :Energy demand is higher and more stable during heat extremes
- Cold periods exhibit greater volatility, likely driven by heterogeneous heating behavior



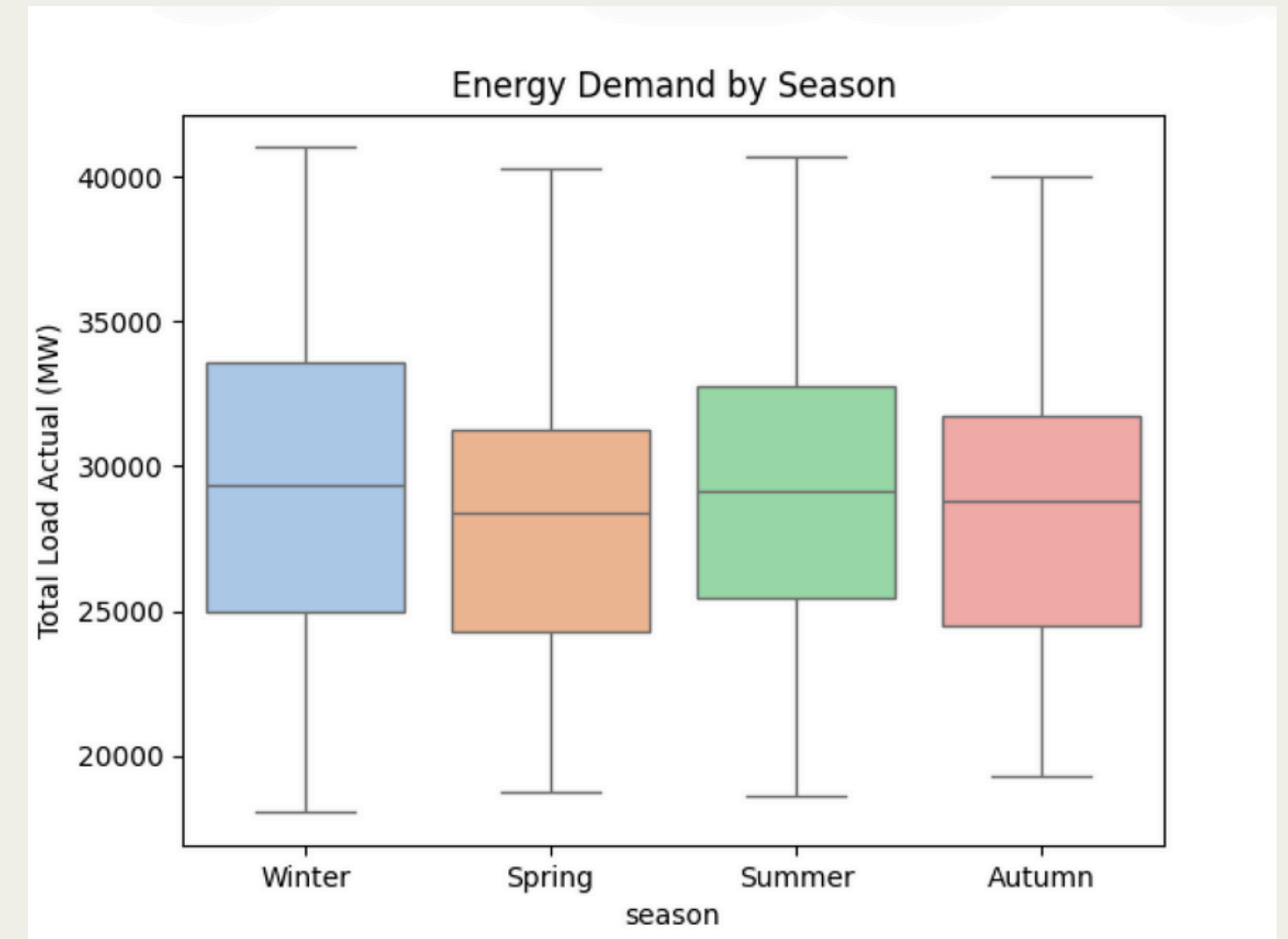
- Shapiro–Wilk normality test results:
- Low Temp: $p = 0.0036$ (non-normal, high volatility)
- High Temp: $p = 0.8782$ (approximately normal)
- Welch's T-test: $t = -2.28$, $p = 0.0265$

STATISTICAL ANALYSIS

Seasonal Variations in Energy Generation

Does energy generation exhibit significant fluctuations across the four seasons?

- H0: Energy generation is constant and does not vary across seasons.
- H1: There is a statistically significant difference in energy generation between at least two seasons.
- Analysis based on ~35,000 hourly observations, grouped by astronomical seasons
- Mean total generation by season (MW):
 - Winter: 29,304 MW (highest)
 - Summer: 29,164 MW
 - Autumn: 28,363 MW
 - Spring: 27,966 MW (lowest)
- Generation variability differs across seasons
 - Highest Std. Dev.: Winter (4,985 MW)
 - Lowest Std. Dev.: Spring (4,262 MW)
- One-Way ANOVA: $F = 174.56$, $p < 0.0001$
 - Strong evidence of differences in mean generation across seasons
- Friedman Test (longitudinal consistency):
 - Statistic = 11.10, $p = 0.0112$
 - Seasonal pattern consistent across years

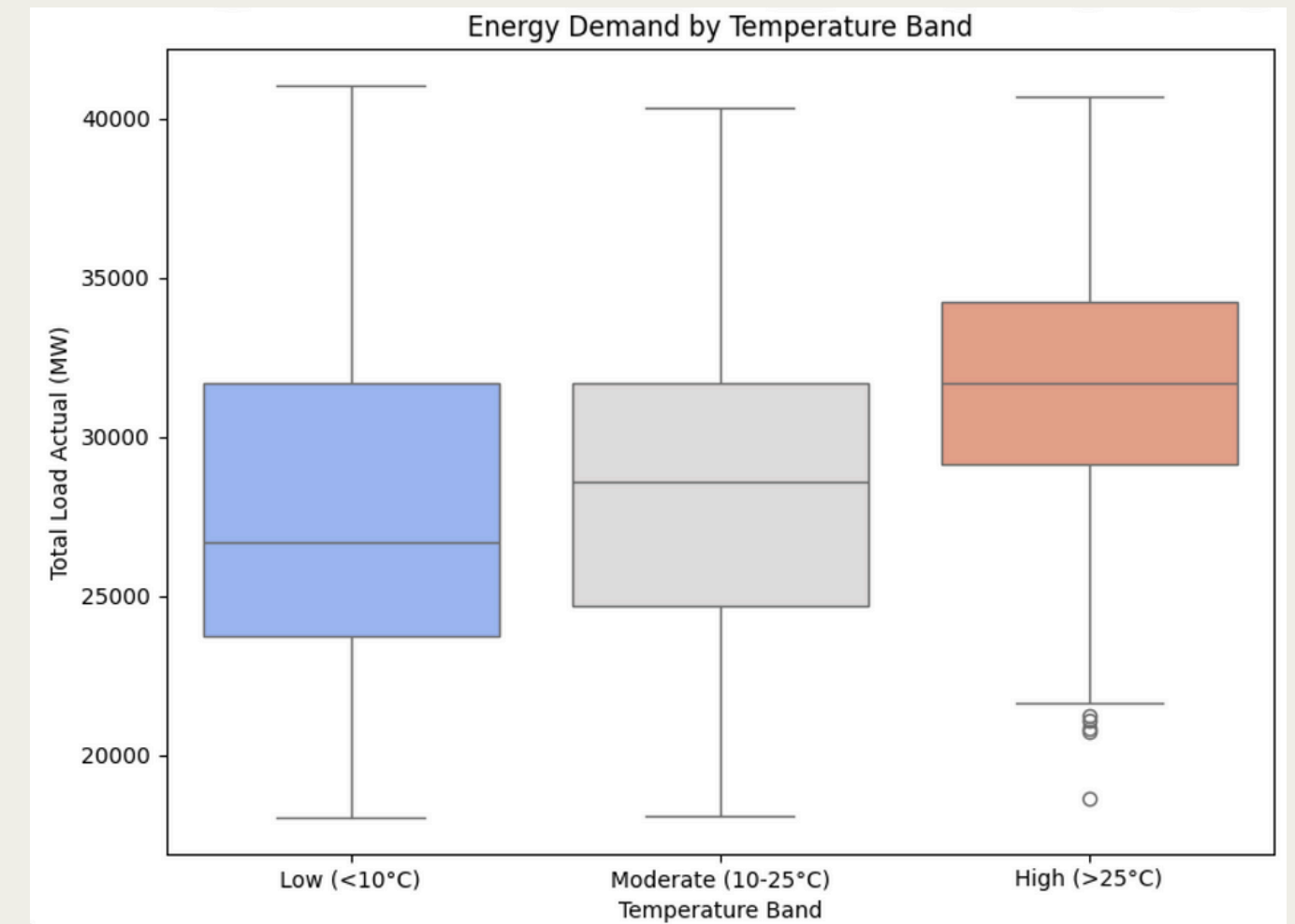


- H0 rejected: Energy generation exhibits statistically significant seasonal cycles
- Winter and summer peaks reflect heating and cooling demand
- Higher winter variance suggests greater sensitivity to extreme weather events

STATISTICAL ANALYSIS

Impact of Temperature Variability on Demand Volatility

- Does temperature variability lead to increased volatility in energy demand, and do extreme temperatures correlate with higher demand fluctuations?
- H0: The variability (standard deviation) of energy demand remains constant across different temperature bands.
- H1: Energy demand variability is significantly higher during temperature extremes compared to moderate periods.
- Demand volatility measured as standard deviation of total load
- Three temperature bands analyzed:
 - Low (< 10°C): 5,059 MW
 - Moderate (10–25°C): 4,351 MW
 - High (> 25°C): 3,617 MW
- H0 rejected : Volatility decreases as temperature increases
- Highest uncertainty observed during cold periods
- Levene's Test: $W = 707.71$, $p < 0.0001$
- Bartlett's Test: $T = 635.12$, $p < 0.0001$
 - Demand variance is not homogeneous across temperature bands

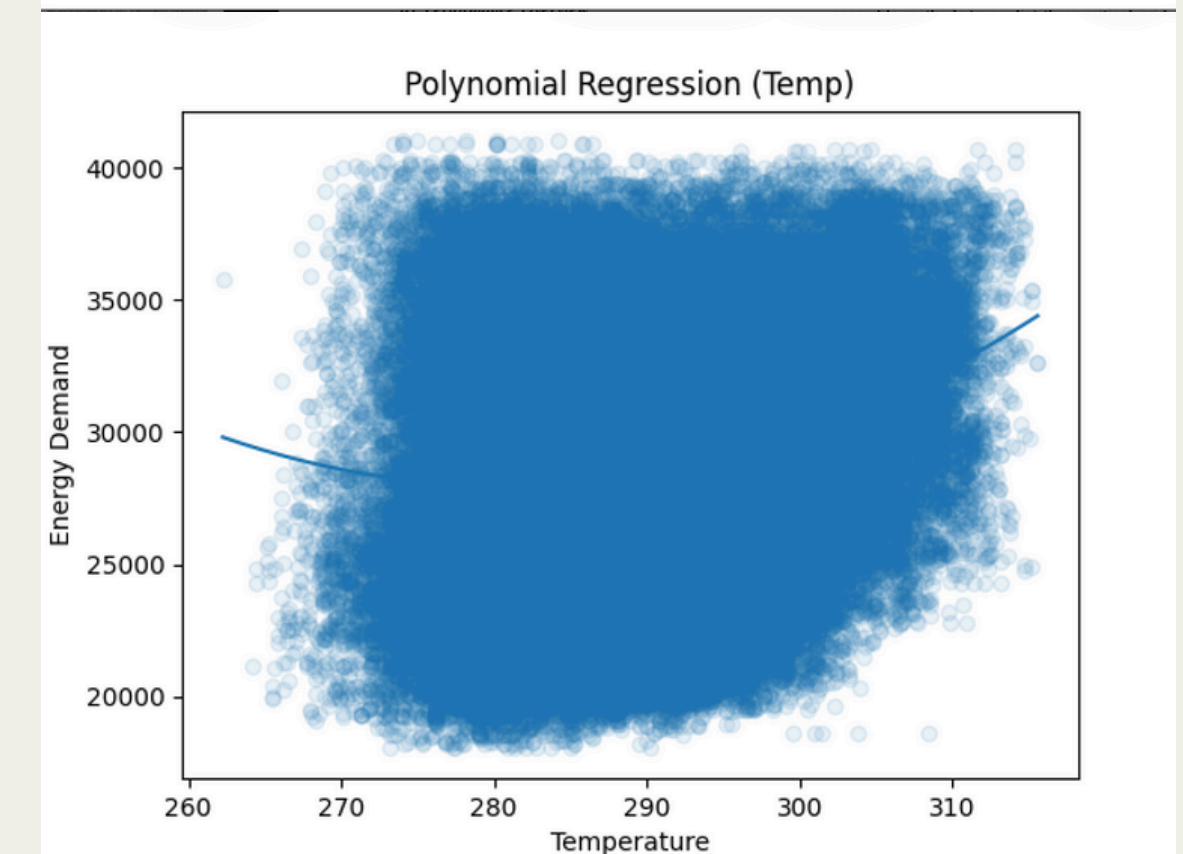
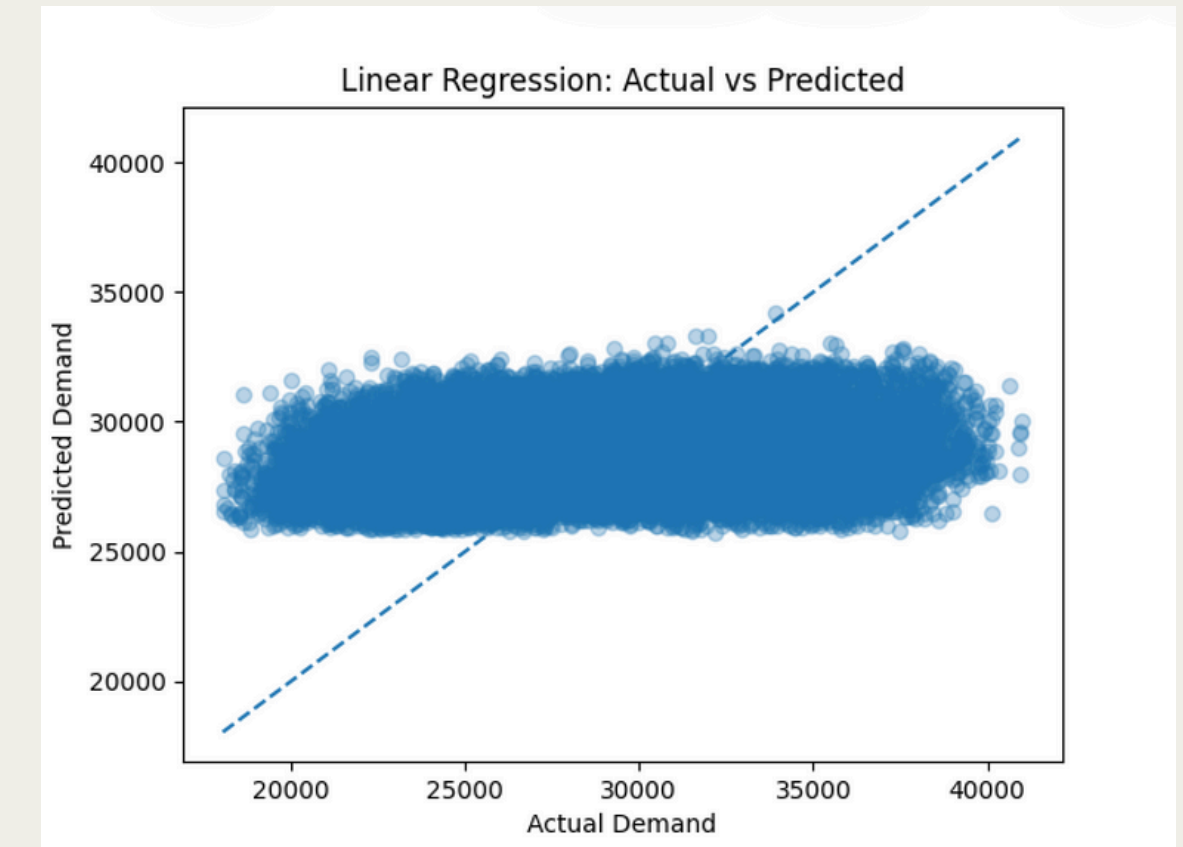


- Extreme low vs high temperatures (mean comparison):
 - T-test: $t = -48.60$, $p < 0.0001$
- Cold temperatures cause nearly 40% higher demand volatility than hot periods
- Heating-related demand is less uniform and more unpredictable than cooling demand

PREDICTIVE MODELING AND REGRESSION ANALYSIS

Baseline Regression Models

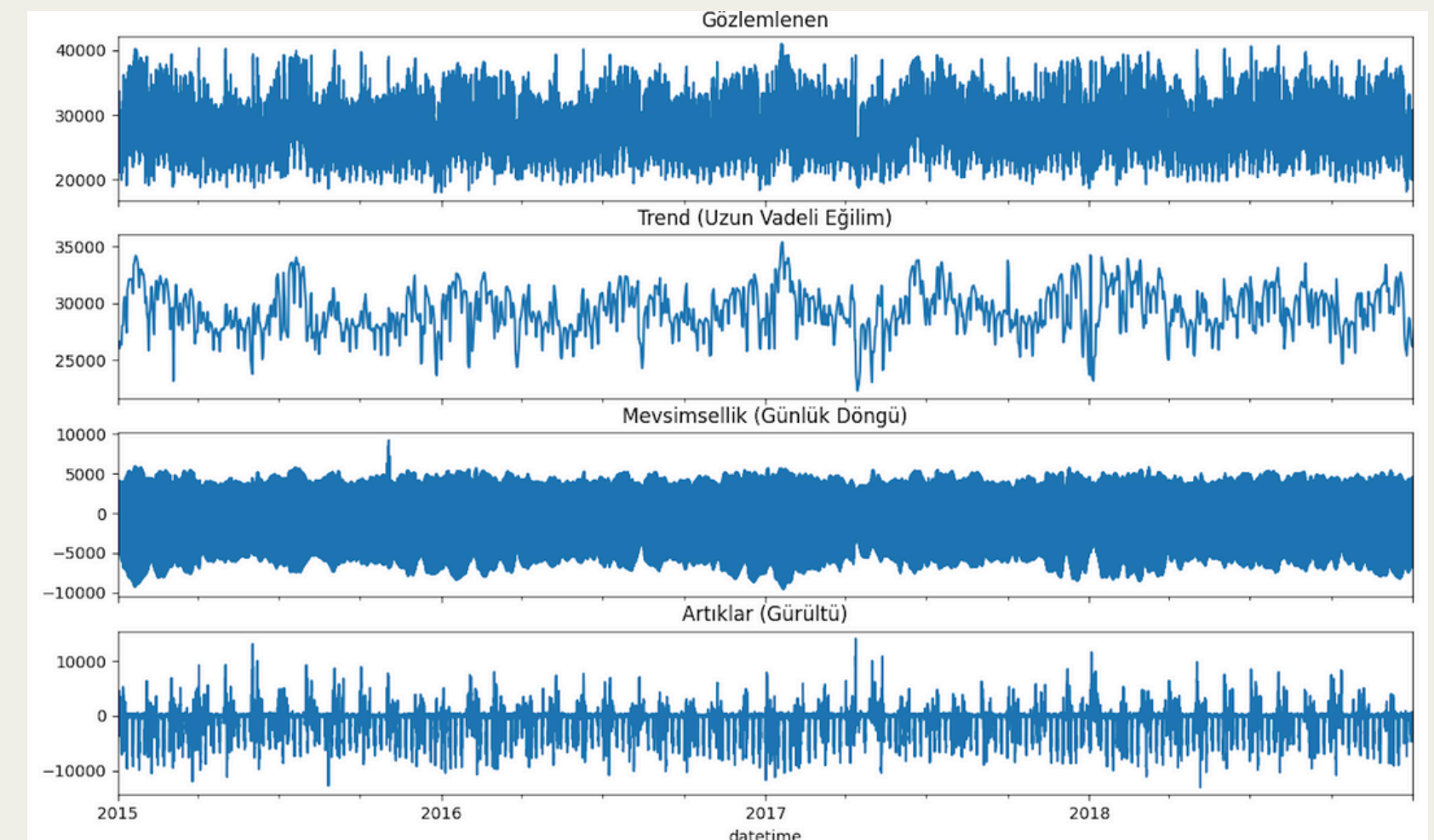
- Baseline models used to link weather variables to electricity demand
- Multivariate OLS regression:
 - Adjusted $R^2 = 0.158 \rightarrow$ weather explains only ~16% of demand variance
 - Durbin-Watson = 0.126 $\rightarrow$ strong autocorrelation (time-series violation)
 - Model statistically significant overall ($F = 1644, p < 0.001$)
- Key coefficients:
 - Humidity: $\beta = -143.9$ ($p < 0.001$)
 - Temperature: $\beta = -50.6$ ($p < 0.001$)
 - Wind speed: insignificant ($p = 0.952$)
- Logistic regression (classification):
 - Binary (High vs Low): 66.0% accuracy
 - Multiclass: 47.5% accuracy, poor “Medium” demand detection ($F1 = 0.24$)
- Polynomial regression (2nd order):
 - $R^2 = 0.055$
 - Confirms non-linearity but offers very low explanatory power



PREDICTIVE MODELING AND REGRESSION ANALYSIS

Time Series Decomposition

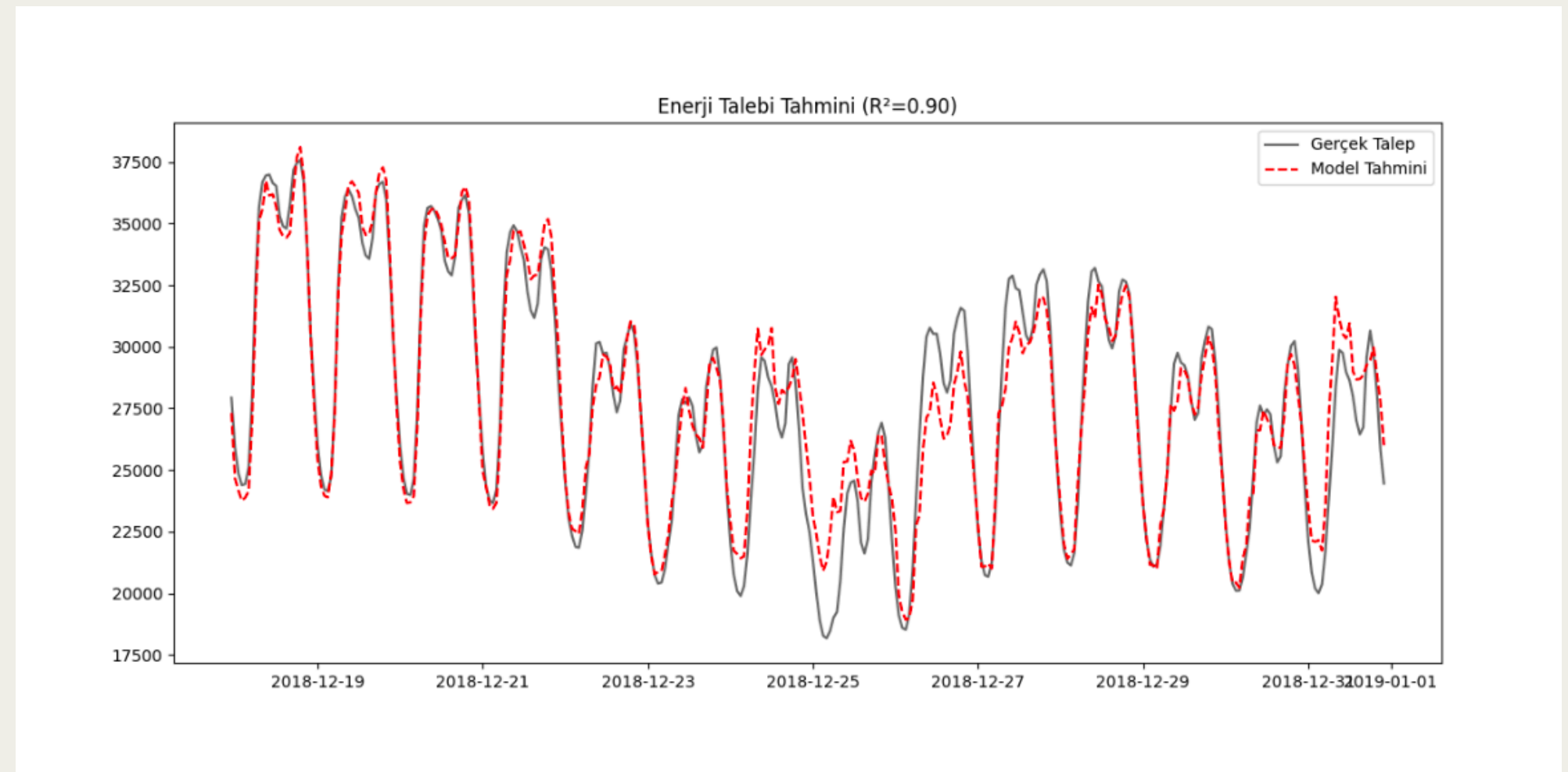
- Seasonal-Trend decomposition (STL) applied to hourly energy demand
- Objective: separate trend, seasonality, and noise components
- Strong daily seasonality clearly observed
- Long-term trend remains relatively stable over 2015–2018
- High noise-to-signal ratio: 0.4893
 - Nearly 50% of demand variation is unexplained by trend and seasonality
- Residual component captures:
 - Sudden demand spikes
 - Extreme weather effects
 - Abrupt changes in industrial consumption
- High residual volatility explains why classical linear models perform poorly



PREDICTIVE MODELING AND REGRESSION ANALYSIS

Gradient Boosting Forecasting-Energy Demand Forecasting

- HistGradientBoosting model trained with:
 - Lagged demand features
 - Rolling statistics
 - Weather variables
- Forecasting performance:
 - $R^2 = 0.8959$ (≈ 0.90 on test data)
- Model accurately captures:
 - Daily demand peaks and troughs
 - Short-term fluctuations despite high noise
- Robust performance despite:
 - High noise-to-signal ratio (0.489)
 - Strong non-linearity in demand patterns
- Significantly outperforms classical time-series models
 - SARIMAX benchmark: $R^2 \approx 0.34$

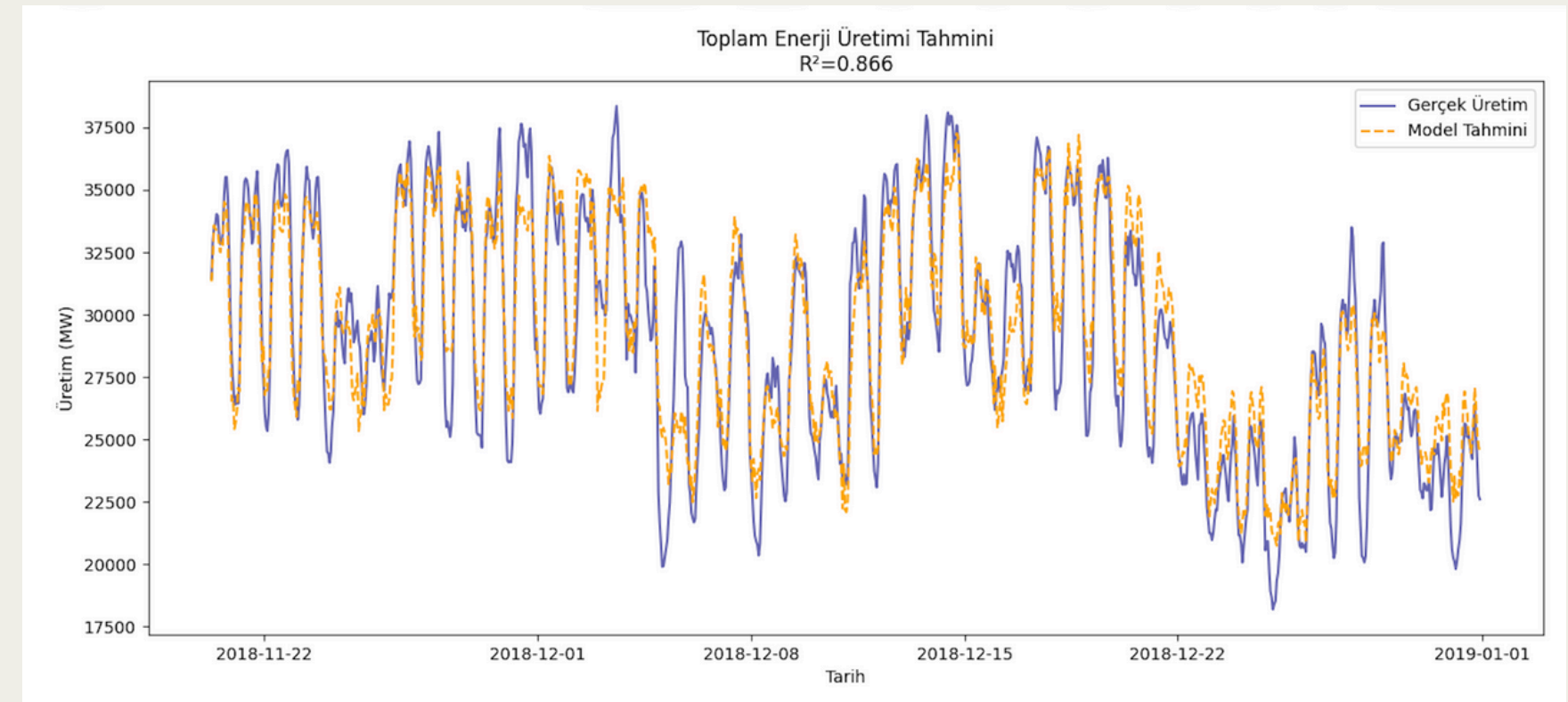


PREDICTIVE MODELING AND REGRESSION ANALYSIS

Gradient Boosting Forecasting

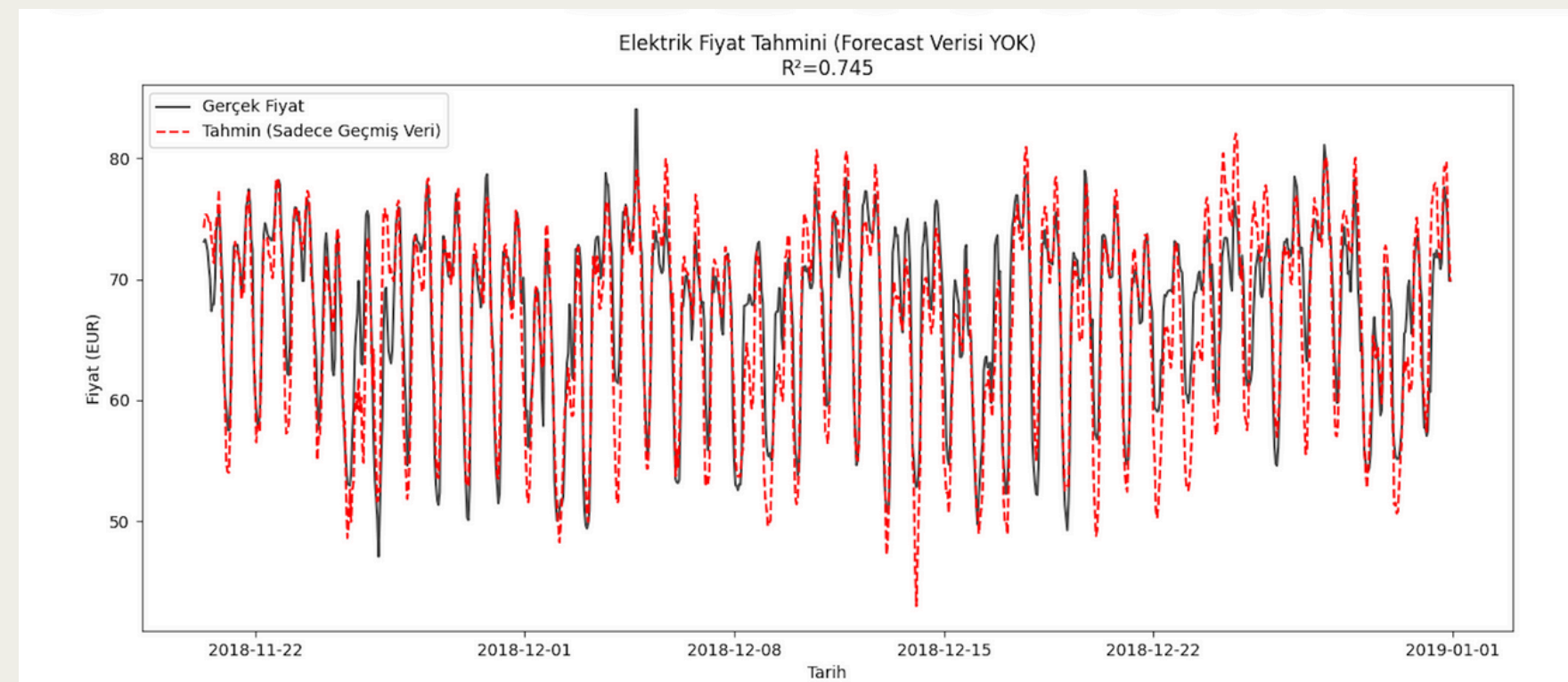
Total Energy Generation Forecasting

- $R^2 = 0.8659$
- MAE = 1,356 MW
- High alignment between predicted and actual generation
- Confirms reliability of:
 - Historical patterns
 - Weather-informed supply-side modeling



Electricity Price Forecasting

- Trained without future weather inputs
- $R^2 = 0.7448$
- Mean error: 2.83 EUR/MWh
- Prices are predictable using:
 - Lagged historical demand
 - Generation dynamics



PREDICTIVE MODELING AND REGRESSION ANALYSIS

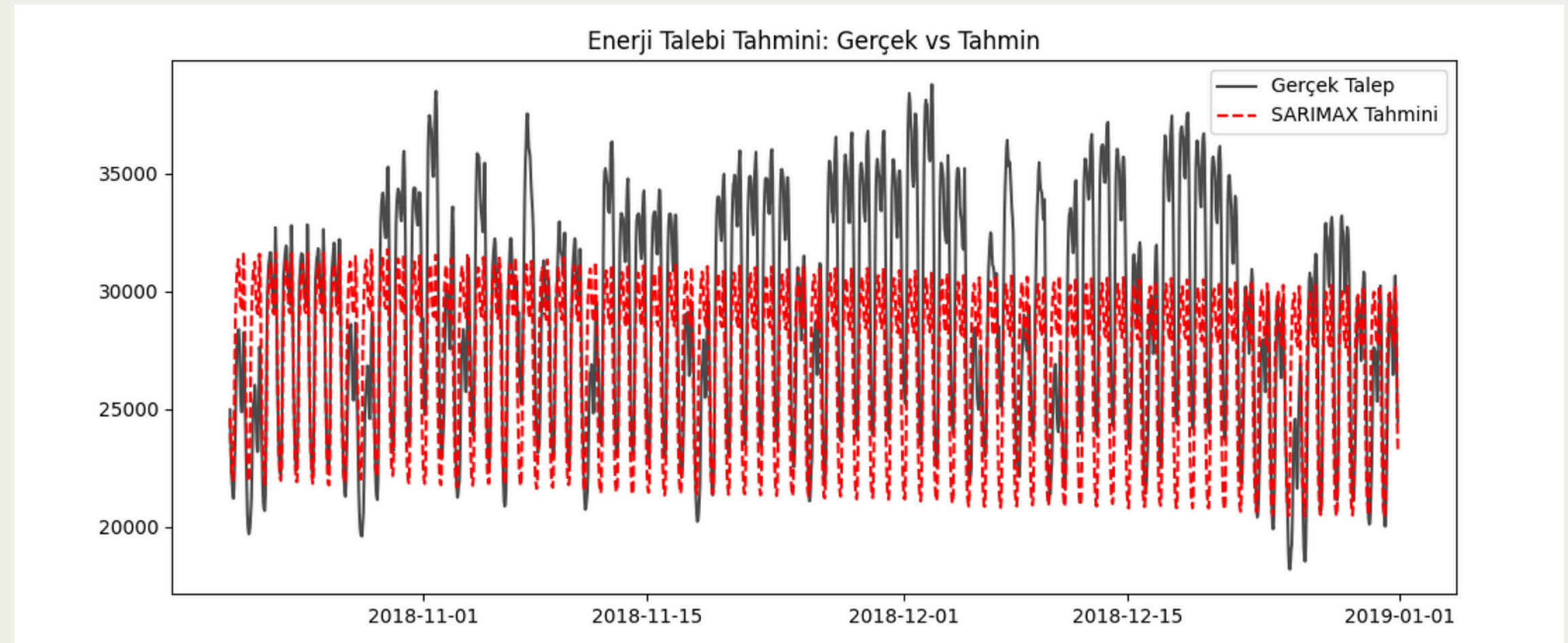
Model Comparison

SARIMAX Model

- $R^2 = 0.3407$
- Failed to converge reliably
- Unable to capture:
 - Complex seasonality
 - Weather-driven variability
 - Non-linear demand dynamics

Gradient Boosting Models

- Demand: $R^2 \approx 0.90$
- Generation: $R^2 \approx 0.87$
- Price: $R^2 \approx 0.74$
- Effectively handles:
 - High-dimensional inputs
 - Non-linear relationships
 - Volatile residual behavior



CONCLUSION

- Weather conditions significantly influence energy systems
 - Solar generation strongly affected by cloud cover and precipitation ($p < 0.0001$)
 - Wind speed alone is insufficient to explain wind generation variability
- Temperature impacts both demand level and demand stability
 - Highest demand volatility observed during cold periods ($< 10^{\circ}\text{C}$)
- Classical statistical models fail under:
 - High volatility
 - Strong non-linearity
 - Temporal dependence
- Gradient Boosting models achieve high predictive accuracy
 - Demand forecasting: $R^2 \approx 0.90$
 - Generation forecasting: $R^2 \approx 0.87$
 - Price forecasting: $R^2 \approx 0.74$

Future Directions

- Develop multivariate wind generation models beyond wind speed
- Apply deep learning approaches (LSTM, Transformer) for complex seasonality
- Incorporate spatial and regional weather effects for improved forecasting